

A knowledge-driven framework for predicting single-cell responses for unprofiled drugs

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A Supplementary Materials

A.1 Detailed experimental results

This section provides the detailed numerical results corresponding to the key findings reported in the main-text Results section. The tables are arranged to follow the order of the Results section. Tables 1 to 3 show results of the zero-shot generalization to unseen cell type-drug combinations (Section 2.2). Tables 4 to 9 show results of the zero-shot generalization to unprofiled drugs (Section 2.3), Table 10 shows the GSEA results of the four prioritized anti-cancer drugs (Section 2.4), and Tables 11 and 12 show analysis of knowledge scale and knowledge type diversity (Section 2.5).

Supplementary Table 1 | Average performance comparison between MAP and baseline models on 6 cell lines selected from Tahoe-100M dataset, under the regime of zero-shot generalization to unseen cell type-drug combinations. Pearson delta correlation, Perturbation discrimination score and direction accuracy are calculated. 95% confidence intervals are reported.

Method	HVG					Top-50 DEG			
	Pearson $\Delta \uparrow$	PDS $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$	Pearson $\Delta \uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$
trainMean	0.331 $\pm$ 0.005	0.024 $\pm$ 0.001	0.553 $\pm$ 0.016	0.0493 $\pm$ 0.0035	–	0.419 $\pm$ 0.004	0.741 $\pm$ 0.014	0.6843 $\pm$ 0.0412	–
PRnet	0.327 $\pm$ 0.004	0.096 $\pm$ 0.003	0.662 $\pm$ 0.017	0.1207 $\pm$ 0.0068	1.7157 $\pm$ 0.0864	0.479 $\pm$ 0.004	0.618 $\pm$ 0.012	0.4689 $\pm$ 0.0237	1.8812 $\pm$ 0.1125
chemCPA	0.677 $\pm$ 0.014	0.285 $\pm$ 0.004	0.636 $\pm$ 0.015	0.0959 $\pm$ 0.0049	0.6548 $\pm$ 0.0393	0.724 $\pm$ 0.021	0.832 $\pm$ 0.013	0.3382 $\pm$ 0.0169	1.1689 $\pm$ 0.0701
XPert	0.707 $\pm$ 0.011	0.706 $\pm$ 0.018	0.630 $\pm$ 0.015	0.0907 $\pm$ 0.0057	0.5453 $\pm$ 0.0247	0.782 $\pm$ 0.018	0.843 $\pm$ 0.013	0.2771 $\pm$ 0.0195	0.8064 $\pm$ 0.0403
CRISP	0.693 $\pm$ 0.015	0.754 $\pm$ 0.028	0.608 $\pm$ 0.005	0.0113 $\pm$ 0.0007	0.1374 $\pm$ 0.0091	0.797 $\pm$ 0.017	0.855 $\pm$ 0.022	0.1515 $\pm$ 0.0104	0.3189 $\pm$ 0.0191
STATE	0.714 $\pm$ 0.012	0.730 $\pm$ 0.021	0.702 $\pm$ 0.016	0.0120 $\pm$ 0.0006	0.1462 $\pm$ 0.0082	0.807 $\pm$ 0.008	0.877 $\pm$ 0.025	0.2094 $\pm$ 0.0118	0.5576 $\pm$ 0.0335
MAP	0.878 $\pm$ 0.019	0.906 $\pm$ 0.013	0.791 $\pm$ 0.029	0.0098 $\pm$ 0.0005	0.0467 $\pm$ 0.0026	0.930 $\pm$ 0.005	0.990 $\pm$ 0.016	0.1166 $\pm$ 0.0072	0.2035 $\pm$ 0.0122

Supplementary Table 2 | Cell type specific analysis of Tahoe-100M dataset, under the regime of zero-shot generalization to unseen cell type-drug combinations. We report cell-line-wise performance (top-50 DEG Pearson delta correlation and direction accuracy) on 6 curated cell lines, averaged across testing drug sets of each cell line.

Model	A-172	A-498	A-549	Hep-G2/C3A	PANC-1	SK-MEL-2	Average
<i>Metric: Pearson Δ top-50 DEG $\uparrow$</i>							
trainMean	0.330 $\pm$ 0.014	0.486 $\pm$ 0.015	0.338 $\pm$ 0.013	0.347 $\pm$ 0.019	0.514 $\pm$ 0.013	0.521 $\pm$ 0.024	0.423 $\pm$ 0.016
PRnet	0.380 $\pm$ 0.003	0.590 $\pm$ 0.004	0.363 $\pm$ 0.023	0.462 $\pm$ 0.003	0.572 $\pm$ 0.003	0.497 $\pm$ 0.024	0.477 $\pm$ 0.010
chemCPA	0.768 $\pm$ 0.020	0.682 $\pm$ 0.034	0.761 $\pm$ 0.019	0.717 $\pm$ 0.021	0.780 $\pm$ 0.018	0.733 $\pm$ 0.015	0.740 $\pm$ 0.021
XPert	0.729 $\pm$ 0.012	0.819 $\pm$ 0.021	0.732 $\pm$ 0.017	0.824 $\pm$ 0.025	0.781 $\pm$ 0.010	0.819 $\pm$ 0.019	0.784 $\pm$ 0.018
CRISP	0.750 $\pm$ 0.009	0.837 $\pm$ 0.021	0.831 $\pm$ 0.018	0.764 $\pm$ 0.020	0.801 $\pm$ 0.027	0.793 $\pm$ 0.013	0.795 $\pm$ 0.018
STATE	0.763 $\pm$ 0.008	0.840 $\pm$ 0.015	0.784 $\pm$ 0.010	0.837 $\pm$ 0.019	0.799 $\pm$ 0.006	0.818 $\pm$ 0.012	0.807 $\pm$ 0.011
MAP	0.957 $\pm$ 0.007	0.945 $\pm$ 0.006	0.915 $\pm$ 0.007	0.895 $\pm$ 0.011	0.935 $\pm$ 0.017	0.925 $\pm$ 0.009	0.929 $\pm$ 0.009
<i>Metric: Direction Acc top-50 DEG $\uparrow$</i>							
trainMean	0.670 $\pm$ 0.011	0.841 $\pm$ 0.019	0.682 $\pm$ 0.016	0.720 $\pm$ 0.019	0.808 $\pm$ 0.015	0.686 $\pm$ 0.007	0.734 $\pm$ 0.014
PRnet	0.553 $\pm$ 0.012	0.678 $\pm$ 0.012	0.503 $\pm$ 0.010	0.584 $\pm$ 0.010	0.688 $\pm$ 0.013	0.700 $\pm$ 0.018	0.617 $\pm$ 0.012
chemCPA	0.804 $\pm$ 0.017	0.872 $\pm$ 0.034	0.840 $\pm$ 0.015	0.809 $\pm$ 0.010	0.851 $\pm$ 0.019	0.838 $\pm$ 0.016	0.835 $\pm$ 0.018
XPert	0.813 $\pm$ 0.018	0.804 $\pm$ 0.007	0.863 $\pm$ 0.016	0.828 $\pm$ 0.013	0.847 $\pm$ 0.009	0.892 $\pm$ 0.022	0.841 $\pm$ 0.014
CRISP	0.822 $\pm$ 0.020	0.908 $\pm$ 0.021	0.869 $\pm$ 0.015	0.832 $\pm$ 0.034	0.879 $\pm$ 0.021	0.868 $\pm$ 0.022	0.863 $\pm$ 0.022
STATE	0.842 $\pm$ 0.018	0.903 $\pm$ 0.031	0.851 $\pm$ 0.024	0.918 $\pm$ 0.036	0.885 $\pm$ 0.012	0.859 $\pm$ 0.027	0.877 $\pm$ 0.025
MAP	0.995 $\pm$ 0.005	0.992 $\pm$ 0.006	0.985 $\pm$ 0.012	0.978 $\pm$ 0.020	0.990 $\pm$ 0.008	0.988 $\pm$ 0.010	0.988 $\pm$ 0.010

Supplementary Table 3 | Cell type specific analysis of OP3 and SciPlex3 datasets, under the regime of zero-shot generalization to unseen cell type-drug combinations. We show fine-grained performance (top-50 DEG Pearson delta correlation) on 6 cell types of OP3 (B cells, Myeloid cells, NK cells, T cells CD4+, T cells CD8+ and T regulatory cells) and 3 cell lines of SciPlex3 (A-549, K-562 and MCF-7, averaged across testing drug sets of each cell type).

	MAP	STATE	XPert	CRISP	chemCPA	PRnet	trainMean
B cells	0.908 ± 0.018	0.696 ± 0.018	0.594 ± 0.009	0.714 ± 0.011	0.690 ± 0.010	0.556 ± 0.023	0.594 ± 0.015
Myeloid cells	0.730 ± 0.007	0.607 ± 0.009	0.590 ± 0.014	0.624 ± 0.021	0.594 ± 0.013	0.492 ± 0.035	0.542 ± 0.015
NK cells	0.809 ± 0.015	0.648 ± 0.014	0.666 ± 0.021	0.673 ± 0.014	0.648 ± 0.012	0.504 ± 0.027	0.560 ± 0.015
T cells CD4+	0.847 ± 0.010	0.638 ± 0.017	0.693 ± 0.018	0.684 ± 0.012	0.671 ± 0.013	0.516 ± 0.031	0.570 ± 0.012
T cells CD8+	0.749 ± 0.009	0.610 ± 0.011	0.622 ± 0.011	0.653 ± 0.011	0.639 ± 0.005	0.506 ± 0.023	0.546 ± 0.016
T regulatory cells	0.857 ± 0.012	0.712 ± 0.022	0.621 ± 0.023	0.697 ± 0.022	0.675 ± 0.014	0.524 ± 0.034	0.579 ± 0.010
Average	0.815 ± 0.014	0.652 ± 0.016	0.631 ± 0.010	0.673 ± 0.009	0.650 ± 0.013	0.524 ± 0.048	0.568 ± 0.018
A-549	0.808 ± 0.009	0.724 ± 0.014	0.694 ± 0.016	0.764 ± 0.018	0.617 ± 0.023	0.503 ± 0.024	0.456 ± 0.006
K-562	0.857 ± 0.006	0.793 ± 0.005	0.772 ± 0.007	0.812 ± 0.011	0.783 ± 0.010	0.574 ± 0.012	0.402 ± 0.004
MCF7	0.769 ± 0.008	0.717 ± 0.009	0.691 ± 0.020	0.615 ± 0.023	0.682 ± 0.009	0.461 ± 0.013	0.441 ± 0.006
Average	0.811 ± 0.008	0.745 ± 0.008	0.719 ± 0.012	0.724 ± 0.015	0.697 ± 0.015	0.512 ± 0.018	0.433 ± 0.006

Supplementary Table 4 | Performance comparison between MAP and baseline models on 6 cell lines selected from Tahoe-100M dataset, perturbed with **seen drugs**. 95% confidence intervals are reported. Best results are highlighted in bold.

Metric	MAP	STATE	CRISP	XPert	chemCPA	PRnet	trainMean
Pearson Δ HVG	0.869 ± 0.023	0.827 ± 0.023	0.721 ± 0.019	0.707 ± 0.022	0.718 ± 0.018	0.262 ± 0.007	0.331 ± 0.009
PDS HVG	0.992 ± 0.026	0.911 ± 0.018	0.790 ± 0.016	0.771 ± 0.014	0.266 ± 0.006	0.057 ± 0.001	0.017 ± 0.001
Direction Acc HVG	0.816 ± 0.019	0.731 ± 0.019	0.710 ± 0.017	0.694 ± 0.008	0.664 ± 0.017	0.637 ± 0.019	0.454 ± 0.006
Pearson Δ top-50 DEG	0.969 ± 0.021	0.943 ± 0.016	0.896 ± 0.023	0.881 ± 0.027	0.711 ± 0.014	0.312 ± 0.010	0.461 ± 0.016
Direction Acc top-50 DEG	0.987 ± 0.013	0.922 ± 0.026	0.842 ± 0.026	0.870 ± 0.011	0.846 ± 0.015	0.565 ± 0.011	0.689 ± 0.020
MSE HVG	0.0062 ± 0.0003	0.0079 ± 0.0004	0.0101 ± 0.0006	0.0412 ± 0.0021	0.0627 ± 0.0032	0.0791 ± 0.0045	0.0421 ± 0.0023
MSE top-50 DE	0.0486 ± 0.0029	0.0617 ± 0.0035	0.1125 ± 0.0068	0.1634 ± 0.0091	0.2146 ± 0.0127	0.4558 ± 0.0264	0.2850 ± 0.0173
Wasserstein HVG	0.0900 ± 0.0045	0.1603 ± 0.0096	0.1150 ± 0.0063	0.1939 ± 0.0134	0.2875 ± 0.0201	0.5694 ± 0.0402	–
Wasserstein top-50 DE	0.1221 ± 0.0067	0.2426 ± 0.0170	0.4438 ± 0.0266	0.4896 ± 0.0392	0.7216 ± 0.0505	1.1198 ± 0.0784	–

Supplementary Table 5 | Performance comparison between MAP and baseline models on 6 cell lines selected from Tahoe-100M dataset, perturbed with **unseen drugs**. 95% confidence intervals are reported. Best results are highlighted in bold.

Metric	MAP	CRISP	XPert	chemCPA	PRnet	trainMean
Pearson Δ HVG	0.659 ± 0.014	0.489 ± 0.012	0.505 ± 0.006	0.202 ± 0.004	0.223 ± 0.005	0.277 ± 0.005
PDS HVG	0.344 ± 0.006	0.175 ± 0.005	0.156 ± 0.025	0.022 ± 0.000	0.003 ± 0.000	0.013 ± 0.000
Direction Acc HVG	0.807 ± 0.021	0.655 ± 0.021	0.630 ± 0.028	0.567 ± 0.009	0.632 ± 0.013	0.470 ± 0.009
Pearson Δ top-50 DEG	0.822 ± 0.023	0.700 ± 0.016	0.704 ± 0.019	0.671 ± 0.015	0.327 ± 0.009	0.428 ± 0.010
Direction Acc top-50 DEG	0.934 ± 0.024	0.724 ± 0.020	0.759 ± 0.013	0.559 ± 0.012	0.531 ± 0.014	0.693 ± 0.013
MSE HVG	0.0097 ± 0.0004	0.0118 ± 0.0005	0.0593 ± 0.0025	0.0862 ± 0.0034	0.1089 ± 0.0068	0.0470 ± 0.0019
MSE top-50 DE	0.1264 ± 0.0046	0.1579 ± 0.0097	0.2431 ± 0.0079	0.2943 ± 0.0118	0.5462 ± 0.0305	0.3340 ± 0.0168
Wasserstein HVG	0.1216 ± 0.0073	0.1465 ± 0.0081	0.6030 ± 0.0362	0.8642 ± 0.0605	1.7726 ± 0.1064	–
Wasserstein top-50 DE	0.2299 ± 0.0138	0.3187 ± 0.0223	0.9862 ± 0.0592	1.8368 ± 0.1102	2.4125 ± 0.1448	–

Supplementary Table 6 | Compound-specific analysis under the regime of zero-shot generalization to unprofiled drugs. Top 50 DE genes’ Pearson delta correlation are reported with the 95% confidence intervals.

Compound	MAP	XPert	CRISP	chemCPA	PRnet	trainMean
Balsalazide (sodium hydrate)	0.854 ± 0.029	0.806 ± 0.031	0.836 ± 0.068	0.575 ± 0.043	0.178 ± 0.486	0.602 ± 0.088
Bergenin	0.802 ± 0.076	0.794 ± 0.045	0.504 ± 0.038	0.556 ± 0.047	0.140 ± 0.251	0.203 ± 0.094
Brivudine	0.678 ± 0.145	0.556 ± 0.072	0.555 ± 0.045	0.594 ± 0.049	0.282 ± 0.206	0.559 ± 0.126
Carbidopa (monohydrate)	0.965 ± 0.033	0.803 ± 0.058	0.884 ± 0.068	0.763 ± 0.052	0.425 ± 0.102	0.382 ± 0.103
Ciclopirox	0.620 ± 0.042	0.528 ± 0.090	0.541 ± 0.034	0.532 ± 0.041	0.222 ± 0.224	0.313 ± 0.104
CP21R7	0.865 ± 0.043	0.742 ± 0.067	0.652 ± 0.049	0.432 ± 0.033	0.130 ± 0.229	0.424 ± 0.108
Drospirenone	0.941 ± 0.058	0.820 ± 0.029	0.882 ± 0.059	0.745 ± 0.053	0.487 ± 0.316	0.242 ± 0.130
ERK5-IN-2	0.869 ± 0.041	0.718 ± 0.084	0.635 ± 0.051	0.743 ± 0.055	0.324 ± 0.219	0.419 ± 0.086
Estrone sulfate (potassium)	0.816 ± 0.064	0.601 ± 0.053	0.706 ± 0.050	0.653 ± 0.026	0.346 ± 0.072	0.260 ± 0.064
Idarubicin (hydrochloride)	0.737 ± 0.072	0.599 ± 0.076	0.458 ± 0.035	0.582 ± 0.043	0.385 ± 0.263	0.297 ± 0.091
Lidocaine (hydrochloride)	0.932 ± 0.034	0.838 ± 0.061	0.690 ± 0.028	0.733 ± 0.057	0.329 ± 0.218	0.282 ± 0.068
Nafamostat (mesylate)	0.808 ± 0.193	0.756 ± 0.038	0.701 ± 0.055	0.700 ± 0.056	0.497 ± 0.080	0.365 ± 0.072
Ralimetinib dimesylate	0.757 ± 0.027	0.695 ± 0.097	0.623 ± 0.049	0.680 ± 0.057	0.612 ± 0.228	0.421 ± 0.072
Sildenafil	0.942 ± 0.038	0.830 ± 0.049	0.852 ± 0.068	0.775 ± 0.055	0.597 ± 0.236	0.412 ± 0.050
Sivelestat (sodium tetrahydrate)	0.916 ± 0.022	0.767 ± 0.082	0.837 ± 0.059	0.789 ± 0.063	0.559 ± 0.240	0.408 ± 0.046
ULK-101	0.713 ± 0.095	0.576 ± 0.034	0.610 ± 0.054	0.572 ± 0.035	0.344 ± 0.221	0.364 ± 0.045

Supplementary Table 7 | Compound-specific analysis under the regime of zero-shot generalization to unprofiled drugs. Top 50 DE genes’ direction accuracy are reported with the 95% confidence intervals.

Compound	MAP	XPert	CRISP	chemCPA	PRnet	trainMean
Balsalazide (sodium hydrate)	0.937 ± 0.025	0.801 ± 0.051	0.605 ± 0.022	0.676 ± 0.047	0.700 ± 0.039	0.815 ± 0.039
Bergenin	0.930 ± 0.049	0.846 ± 0.036	0.876 ± 0.035	0.737 ± 0.046	0.526 ± 0.028	0.616 ± 0.088
Brivudine	0.841 ± 0.066	0.834 ± 0.073	0.839 ± 0.062	0.820 ± 0.070	0.622 ± 0.043	0.868 ± 0.090
Carbidopa (monohydrate)	0.959 ± 0.025	0.839 ± 0.088	0.728 ± 0.054	0.715 ± 0.052	0.468 ± 0.029	0.673 ± 0.070
Ciclopirox	0.862 ± 0.065	0.768 ± 0.078	0.636 ± 0.044	0.390 ± 0.035	0.524 ± 0.033	0.739 ± 0.065
CP21R7	0.938 ± 0.048	0.905 ± 0.081	0.851 ± 0.062	0.762 ± 0.072	0.582 ± 0.027	0.680 ± 0.050
Drospirenone	0.968 ± 0.029	0.652 ± 0.056	0.688 ± 0.056	0.626 ± 0.046	0.475 ± 0.070	0.533 ± 0.114
ERK5-IN-2	0.928 ± 0.034	0.815 ± 0.067	0.895 ± 0.065	0.544 ± 0.041	0.595 ± 0.037	0.572 ± 0.069
Estrone sulfate (potassium)	0.971 ± 0.032	0.789 ± 0.029	0.783 ± 0.063	0.720 ± 0.068	0.495 ± 0.023	0.744 ± 0.040
Idarubicin (hydrochloride)	0.944 ± 0.021	0.597 ± 0.034	0.782 ± 0.052	0.441 ± 0.031	0.567 ± 0.027	0.421 ± 0.048
Lidocaine (hydrochloride)	0.970 ± 0.028	0.862 ± 0.059	0.736 ± 0.058	0.708 ± 0.045	0.527 ± 0.023	0.732 ± 0.062
Nafamostat (mesylate)	0.826 ± 0.051	0.619 ± 0.041	0.445 ± 0.030	0.758 ± 0.028	0.533 ± 0.039	0.676 ± 0.078
Ralimetinib dimesylate	0.889 ± 0.046	0.583 ± 0.027	0.485 ± 0.031	0.682 ± 0.057	0.566 ± 0.056	0.735 ± 0.068
Sildenafil	0.985 ± 0.030	0.666 ± 0.062	0.907 ± 0.049	0.782 ± 0.053	0.492 ± 0.082	0.606 ± 0.036
Sivelestat (sodium tetrahydrate)	0.969 ± 0.065	0.784 ± 0.083	0.800 ± 0.031	0.693 ± 0.052	0.496 ± 0.026	0.736 ± 0.038
ULK-101	0.896 ± 0.037	0.807 ± 0.045	0.746 ± 0.069	0.539 ± 0.044	0.592 ± 0.075	0.821 ± 0.054

Supplementary Table 8 | Average performance comparison between MAP and baseline models on OP3 dataset, under the regime of zero-shot generalization to unprofiled drugs. Pearson delta correlation, perturbation discrimination score and direction accuracy are reported.

Method	HVG					Top-50 DEG			
	Pearson $\Delta \uparrow$	PDS $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$	Pearson $\Delta \uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$
trainMean	0.239 $\pm$ 0.015	0.016 $\pm$ 0.001	0.419 $\pm$ 0.009	1.5631 $\pm$ 0.0153	–	0.468 $\pm$ 0.009	0.658 $\pm$ 0.013	2.8660 $\pm$ 0.0397	–
PRnet	0.246 $\pm$ 0.008	0.048 $\pm$ 0.003	0.456 $\pm$ 0.013	2.0570 $\pm$ 0.0657	1.1050 $\pm$ 0.0497	0.415 $\pm$ 0.012	0.686 $\pm$ 0.013	4.6733 $\pm$ 0.1685	2.3274 $\pm$ 0.1396
chemCPA	0.473 $\pm$ 0.008	0.090 $\pm$ 0.003	0.536 $\pm$ 0.014	0.4878 $\pm$ 0.0416	0.5937 $\pm$ 0.0356	0.663 $\pm$ 0.009	0.783 $\pm$ 0.014	1.8259 $\pm$ 0.0656	1.2381 $\pm$ 0.0743
CRISP	0.525 $\pm$ 0.012	0.106 $\pm$ 0.005	0.571 $\pm$ 0.017	0.2214 $\pm$ 0.0183	0.5432 $\pm$ 0.0326	0.652 $\pm$ 0.014	0.815 $\pm$ 0.020	1.0321 $\pm$ 0.0513	0.7727 $\pm$ 0.0386
XPert	0.473 $\pm$ 0.012	0.125 $\pm$ 0.028	0.532 $\pm$ 0.007	0.1232 $\pm$ 0.0102	0.2781 $\pm$ 0.0195	0.631 $\pm$ 0.031	0.821 $\pm$ 0.019	1.3297 $\pm$ 0.0921	0.6517 $\pm$ 0.0456
MAP	0.601 $\pm$ 0.021	0.263 $\pm$ 0.009	0.692 $\pm$ 0.019	0.0288 $\pm$ 0.0031	0.2229 $\pm$ 0.0134	0.861 $\pm$ 0.007	0.921 $\pm$ 0.014	0.6176 $\pm$ 0.0664	0.5986 $\pm$ 0.0359

Supplementary Table 9 | Average performance comparison between MAP and baseline models on SciPlex3 dataset, under the regime of zero-shot generalization to unprofiled drugs. Pearson delta correlation, perturbation discrimination score and direction accuracy are reported.

Method	HVG					Top-50 DEG			
	Pearson $\Delta \uparrow$	PDS $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$	Pearson $\Delta \uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$
trainMean	0.262 $\pm$ 0.006	0.030 $\pm$ 0.018	0.538 $\pm$ 0.011	0.1095 $\pm$ 0.0016	–	0.475 $\pm$ 0.007	0.638 $\pm$ 0.015	0.7117 $\pm$ 0.0356	–
PRnet	0.429 $\pm$ 0.015	0.064 $\pm$ 0.010	0.736 $\pm$ 0.006	0.1527 $\pm$ 0.0108	0.4728 $\pm$ 0.0284	0.601 $\pm$ 0.014	0.707 $\pm$ 0.015	0.8212 $\pm$ 0.0619	1.4895 $\pm$ 0.0894
chemCPA	0.598 $\pm$ 0.009	0.107 $\pm$ 0.027	0.754 $\pm$ 0.023	0.0951 $\pm$ 0.0069	0.2562 $\pm$ 0.0154	0.630 $\pm$ 0.028	0.742 $\pm$ 0.022	0.6410 $\pm$ 0.0410	0.9471 $\pm$ 0.0568
CRISP	0.631 $\pm$ 0.009	0.123 $\pm$ 0.009	0.789 $\pm$ 0.028	0.0569 $\pm$ 0.0032	0.2562 $\pm$ 0.0154	0.723 $\pm$ 0.011	0.803 $\pm$ 0.015	0.5316 $\pm$ 0.0266	0.3698 $\pm$ 0.0222
XPert	0.647 $\pm$ 0.024	0.103 $\pm$ 0.010	0.749 $\pm$ 0.035	0.0345 $\pm$ 0.0021	0.0679 $\pm$ 0.0041	0.694 $\pm$ 0.016	0.816 $\pm$ 0.022	0.2637 $\pm$ 0.0127	0.2906 $\pm$ 0.0203
MAP	0.698 $\pm$ 0.010	0.275 $\pm$ 0.009	0.848 $\pm$ 0.004	0.0294 $\pm$ 0.0018	0.0135 $\pm$ 0.0009	0.827 $\pm$ 0.010	0.913 $\pm$ 0.010	0.1844 $\pm$ 0.0092	0.0632 $\pm$ 0.0038

A.2 Drug combination perturbation prediction

Drug combination perturbation prediction is a practically relevant task, as it corresponds to combination therapies in real-world settings. To evaluate the performance of MAP on this task, we conduct experiments on the Combosciplex [25] dataset, which profiles single-cell responses to both single-drug and two-drug combination treatments. We train MAP on single-drug perturbations and evaluate it on two-drug combinations under three increasingly challenging splits: (i) both drugs are seen during training, (ii) one drug is seen and the other is unseen, and (iii) both drugs are unseen. We compare against two strong baselines commonly used in combinatorial perturbation studies: **Control**, which uses the unperturbed cell expression as the prediction, and **PerturbMean**, which uses the average of training perturbations as the prediction.

As shown in Table 13, MAP consistently outperforms both baseline methods across all three settings and all evaluation metrics. Notably, from the 2 drugs seen setting to the 0 drugs unseen setting, MAP exhibits a more gradual performance degradation than the baseline methods on most metrics (except for the top-50 DEG Direction Accuracy). This trend indicates that MAP provides stronger support for drug generalization. These results demonstrate that MAP generalizes effectively to combinatorial perturbations, including the strict zero-shot case where neither drug has been observed during training.

A.3 Hyperparameter sensitivity analysis

To demonstrate the robustness of our method to parameter choices, we conduct ablation studies on three datasets over several key model hyper-parameters (Tables 14 to 16). Specifically, we examine the hidden dimension of the Transformer encoder, the weight λ of the expression reconstruction loss term in the loss function, and the number of layers in the Transformer encoder. For each hyperparameter, we vary its value around the default setting while keeping all other factors fixed, and analyze the sensitivity of all evaluation metrics to these variations.

Across all three datasets, MAP demonstrates stable performance under a wide range of hyperparameter settings, indicating strong robustness to parameter choices. For the hidden dimension, we observe no clear monotonic trend, and performance remains largely consistent once a moderate capacity is reached, suggesting that the model does not rely on over-parameterization. Similarly, varying the weight λ of the reconstruction loss leads to only marginal changes across metrics, indicating that the model is not sensitive to the exact balance between objectives.

In contrast, increasing the number of Transformer layers from 2 to 6 leads to a consistent but modest improvement in correlation- and direction-based metrics (e.g., Pearson Δ and Direction Accuracy) across all datasets, suggesting that model depth contributes positively to capturing perturbation patterns. However, this improvement is gradual rather than sharp, and shallow configurations already achieve competitive performance.

Importantly, these observations are consistent across different evaluation metrics, including correlation-based and distribution-based measures, and no conflicting trends are observed. Moreover, the default settings consistently lie within a stable performance region rather than at a sharp optimum. Overall, these results demonstrate that MAP achieves strong performance without requiring delicate hyperparameter tuning.

A.4 Ablation study on representation integration

As mentioned in 4.4, MAP integrates three heterogeneous modalities—cell features, gene features, and drug features. These features are concatenated into a token sequence and processed by a Transformer encoder to produce the final representations. This sequence formulation enables cross-modal interactions through self-attention: tokens from different modalities can attend to each other and exchange information. In particular, drug tokens enriched with mechanistic knowledge can interact with gene tokens, allowing the model to capture drug–gene interaction patterns. Meanwhile, the cell token, which is pre-trained to decode jointly with gene-level signals, can aggregate gene-level perturbation information into a compact cell-level representation.

To evaluate the effectiveness of this design, we conduct ablation studies on OP3 under the zero-shot generalization to unprofiled drugs setting. In each variant, we remove one type of feature while keeping the rest unchanged. The drug token is not ablated, as it is an essential input; its role is analyzed in Section 2.5. Specifically, we consider the following four variants: (i) **w/o cell emb**, where the cell embedding is replaced with a randomly initialized token; (ii) **w/o SE & KG gene emb**, where all gene features are removed; (iii) **w/o SE gene emb**, where gene features from the SE encoder are removed; (iv) **w/o KG gene emb**, where knowledge-pretrained gene features are removed.

The results are shown in Supplementary Table 17. Removing the cell feature leads to the largest performance drop, indicating its critical role in aggregating gene-level information. Removing all gene features results in the second largest degradation, highlighting the importance of fine-grained interactions between drug and gene representations, especially those informed by knowledge. Removing either SE gene features or knowledge-pretrained gene features also causes clear performance declines. This can be attributed to their complementary roles: SE gene features preserve a representation space aligned with the cell feature through joint pretraining, facilitating effective aggregation, while knowledge-pretrained gene features encode biological mechanisms that enhance interactions with drug features. As case-level examples, we further examined **cell-line-level results** in the unprofiled-drug settings of OP3 and SciPlex3. For each dataset, we selected the best-performing and worst-performing cell lines based on the overall performance of the full model, resulting in four representative cell-line examples. As shown in Supplementary Table 18, across all four examples, removing the knowledge-pretrained gene features led to consistent performance degradation when considering the full set of evaluation metrics. In summary, the integration of cell, gene, and drug features via a unified attention-based sequence is essential for capturing cross-modal dependencies and achieving strong predictive performance.

A.5 Implementation details and computational cost

The MAP model was implemented using PyTorch(v.2.1) as the deep learning framework. Data preprocessing was conducted with Scanpy(v.1.9) and anndata(v.0.9). The Knowledge-driven multi-modal pre-training was implemented on 4 NVIDIA A-100 GPU. Pre-training on the full MAP-KG took approximately 25h to converge. A scalability analysis of running time is conducted by varying the proportion of training data from 25% to 100%. As shown in Table 19, the convergence time increases approximately linearly with the data size, rising from 6.72 hours at 25% to 25.44 hours at full scale, with the normalized training time (hours per 1% of data) remains nearly constant (0.254–0.269 h/%), indicating reliable extension to larger knowledge graphs without incurring disproportionate computational overhead.

The perturbation response prediction training was implemented on four NVIDIA A100 GPUs. As shown in

Table 20, training on the Tahoe-100M dataset took approximately 11.9 h to converge. We compared the computational cost of MAP with two strong baseline methods, XPert and CRISP. Under the same experimental setting, the convergence time of MAP (11.91 h) was comparable to XPert (12.25 h) and moderately higher than CRISP (8.63 h), indicating improved predictive performance without a substantial increase in training time. In terms of memory usage, MAP required a higher peak single-GPU memory footprint than XPert and CRISP. This is mainly attributable to our representation design, where a relatively long sequence of gene-level tokens is incorporated into the Transformer encoder to enable fine-grained interactions among cell, gene, and drug representations.

Together, these runtime and memory measurements provide a practical reference for estimating the computational resources required to apply MAP to new datasets under comparable model and data settings. In general, full training or large-scale adaptation of MAP is expected to require multi-GPU acceleration with high-memory devices, whereas applying the released pretrained knowledge encoders can reduce the requirement to dataset-specific response-prediction training or fine-tuning.

Supplementary Table 10 | GSEA Results for Four Drugs on A-549 Cell Line. Normalized Enrichment Scores (NES) and False Discovery Rate-adjusted q value (FDR-q) of predictions and values are reported for each gene set (pathway). For every drug, 12 representative gene sets that are linked to each compound's known mechanism of action and to disease-relevant699 programs in A549

Drug	Pathway Name	True NES	True FDR-q	Pred NES	Pred FDR-q
Afatinib	NF- κ B Signaling	1.991	0.199	1.451	0.210
	LPS Inflammatory Response	1.913	0.031	1.175	0.041
	Apoptosis (WP)	1.586	0.091	1.312	0.194
	MYC E-box Targets	1.608	0.188	1.399	0.232
	NF- κ B Fibroblast Targets	1.752	0.417	1.467	0.601
	EGF/EGFR Signaling (WP)	-0.980	0.120	-1.294	0.190
	EGF Response (HeLa)	-0.827	0.125	-0.766	0.230
	ERBB2 Isoform-1 Metastasis	-0.786	0.100	-1.166	0.120
	ERBB2 611CTF Targets	-0.959	0.200	-0.757	0.280
	VEGF-A/VEGFR2 (WP)	-0.973	0.150	-0.899	0.230
	HGF Signaling (AKT1-indep)	-1.307	1.000	-1.337	1.000
	mTOR Signaling Down	-1.336	1.000	-0.675	1.000
Osimertinib	EGFR Signaling 24h Up	1.761	0.251	1.535	0.091
	KRAS Lung Cancer Up	1.987	0.125	2.066	0.005
	MYC Regulated Down	2.084	0.054	2.536	0.002
	NF- κ B Fibroblast Targets	1.686	0.286	2.224	0.003
	Apoptosis (Reactome)	1.943	0.143	1.443	0.579
	Hypoxia Up	1.986	0.119	2.215	0.003
	EGF Interval Up	-0.819	0.100	-0.731	0.090
	Cell Cycle S Phase	-0.650	0.130	-0.981	0.140
	Cell Cycle EG1 Peaks	-0.651	0.100	-1.211	0.170
	IFN- α/β Signaling	-0.885	0.050	-2.104	0.046
	MAPK Signaling (KEGG)	1.011	0.089	2.169	0.003
	PI3K-AKT (WP)	1.240	0.030	2.538	0.002
Crizotinib	NF- κ B Keratinocyte Targets	2.006	0.020	2.326	0.002
	NRF2 Up	2.137	0.018	2.394	0.001
	MYC Bound Genes	2.452	0.003	2.244	0.001
	P53 Network (WP)	2.299	0.007	1.704	0.039
	Cell Cycle (Reactome)	2.104	0.016	2.217	0.001
	MET Signaling (Reactome)	-0.749	0.100	-0.798	0.100
	MET Motility (Reactome)	-0.772	0.130	-1.098	0.160
	EGF/EGFR Signaling (WP)	-0.780	0.090	-0.731	0.130
	STAT3 Targets Down	-1.475	0.300	-1.644	0.484
	ERBB Signaling (WP)	0.984	0.605	1.497	0.105
	PI3K-AKT (Reactome)	1.372	0.256	1.327	0.201
	PI3K-AKT (WP)	1.529	0.151	1.752	0.029
Adagrasib	KRAS Lung Cancer Up	-1.707	0.083	-1.567	0.066
	Hypoxia Up	-2.374	0.005	-2.028	0.028
	Doxorubicin Apoptosis Down	-2.011	0.030	-1.964	0.036
	EMT-like Breast Cancer	-2.224	0.015	-1.855	0.063
	MYC Oncogenic Targets	-1.851	0.051	-2.166	0.026
	MAPK Signaling (KEGG)	-1.177	0.357	-0.958	0.398
	MAPK Signaling (WP)	-1.317	0.257	-1.010	0.555
	RAS Signaling (WP)	-1.621	0.116	-0.996	0.165
	EGFR TKI Resistance (WP)	-1.616	0.117	-1.556	0.172
	PI3K-AKT Cancer Signaling	-1.503	0.160	-1.174	0.311
	TCF-dependent WNT	0.789	0.160	0.764	0.180
	ERBB2/CDC25A Tumorigenesis	0.707	0.160	1.283	0.200

Supplementary Table 11 | Performance on the regime of zero-shot generalization to unprofiled drugs, as we progressively increase the injected knowledge sources. Knowledge variants are defined as: no knowledge, randomly initialized knowledge encoders; +MoleculeSTM, initialization from a pretrained MoleculeSTM checkpoint; +PrimeKG, further pre-training with PrimeKG entities; +MAP-KG, additional pre-training with MAP-KG entities.

Variant	HVG					Top-50 DEG			
	Pearson Δ $\uparrow$	PDS $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$	Pearson Δ $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$
no KG	0.196 $\pm$ 0.009	0.055 $\pm$ 0.003	0.556 $\pm$ 0.012	0.0816 $\pm$ 0.0041	0.3389 $\pm$ 0.0169	0.321 $\pm$ 0.011	0.553 $\pm$ 0.015	1.6240 $\pm$ 0.0893	0.4872 $\pm$ 0.0292
+MolSTM	0.484 $\pm$ 0.011	0.136 $\pm$ 0.005	0.694 $\pm$ 0.016	0.0399 $\pm$ 0.0022	0.2043 $\pm$ 0.0123	0.698 $\pm$ 0.015	0.831 $\pm$ 0.019	0.2081 $\pm$ 0.0125	0.3194 $\pm$ 0.0192
+PrimeKG	0.591 $\pm$ 0.013	0.301 $\pm$ 0.007	0.742 $\pm$ 0.018	0.0135 $\pm$ 0.0007	0.1854 $\pm$ 0.0093	0.708 $\pm$ 0.017	0.862 $\pm$ 0.022	0.2273 $\pm$ 0.0152	0.2371 $\pm$ 0.0142
+MAP-KG	0.659 $\pm$ 0.014	0.344 $\pm$ 0.006	0.807 $\pm$ 0.021	0.0097 $\pm$ 0.0004	0.1216 $\pm$ 0.0073	0.822 $\pm$ 0.023	0.934 $\pm$ 0.024	0.1264 $\pm$ 0.0046	0.2299 $\pm$ 0.0138

Supplementary Table 12 | Ablation of knowledge types by removing one type of knowledge entities from MAP-KG at a time, including drug-side textual attributes (*text*: name/MOA/therapeutic description), gene-side attributes (*seq*: amino-acid sequence; *text*: functional annotations), and relational edges (drug-gene mechanistic interactions and gene-gene functional/physical associations), under the regime of zero-shot generalization to unprofiled drugs.

Variant	HVG					Top-50 DEG			
	Pearson Δ $\uparrow$	PDS $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$	Pearson Δ $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$
w/o gene text	0.594 $\pm$ 0.018	0.271 $\pm$ 0.026	0.735 $\pm$ 0.018	0.0139 $\pm$ 0.0008	0.1765 $\pm$ 0.0106	0.690 $\pm$ 0.013	0.870 $\pm$ 0.010	0.2025 $\pm$ 0.0101	0.2987 $\pm$ 0.0179
w/o gene-gene edges	0.610 $\pm$ 0.027	0.292 $\pm$ 0.019	0.740 $\pm$ 0.020	0.0108 $\pm$ 0.0005	0.1931 $\pm$ 0.0116	0.709 $\pm$ 0.023	0.881 $\pm$ 0.024	0.1342 $\pm$ 0.0083	0.2533 $\pm$ 0.0152
w/o drug text	0.567 $\pm$ 0.025	0.216 $\pm$ 0.023	0.733 $\pm$ 0.002	0.0189 $\pm$ 0.0012	0.1493 $\pm$ 0.0082	0.704 $\pm$ 0.010	0.868 $\pm$ 0.011	0.1587 $\pm$ 0.0076	0.2816 $\pm$ 0.0169
w/o drug-gene edges	0.558 $\pm$ 0.009	0.231 $\pm$ 0.017	0.719 $\pm$ 0.014	0.0176 $\pm$ 0.0010	0.2258 $\pm$ 0.0135	0.679 $\pm$ 0.012	0.854 $\pm$ 0.012	0.2448 $\pm$ 0.0142	0.4129 $\pm$ 0.0248
MAP-KG	0.659 $\pm$ 0.014	0.344 $\pm$ 0.006	0.807 $\pm$ 0.021	0.0097 $\pm$ 0.0004	0.1216 $\pm$ 0.0073	0.822 $\pm$ 0.023	0.934 $\pm$ 0.024	0.1264 $\pm$ 0.0046	0.2299 $\pm$ 0.0138

Supplementary Table 13 | Performance comparison between MAP and baseline models on Combosciplex dataset. Evaluation is conducted on three data splits corresponding to: (i) both drugs seen during training, (ii) one drug seen and one unseen, and (iii) both drugs unseen.

Variant	HVG					Top-50 DEG			
	Pearson Δ $\uparrow$	PDS $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$	Pearson Δ $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$
<i>2 drugs seen</i>									
Control	0.441 $\pm$ 0.015	0.093 $\pm$ 0.004	0.648 $\pm$ 0.026	0.0311 $\pm$ 0.0013	0.3473 $\pm$ 0.0134	0.516 $\pm$ 0.021	0.810 $\pm$ 0.035	0.3418 $\pm$ 0.0134	0.7604 $\pm$ 0.0318
PerturbMean	0.642 $\pm$ 0.032	0.144 $\pm$ 0.006	0.710 $\pm$ 0.029	0.0246 $\pm$ 0.0012	-	0.863 $\pm$ 0.043	0.939 $\pm$ 0.045	0.2097 $\pm$ 0.0097	-
MAP	0.658 $\pm$ 0.026	0.248 $\pm$ 0.012	0.732 $\pm$ 0.037	0.0217 $\pm$ 0.0010	0.1033 $\pm$ 0.0059	0.875 $\pm$ 0.036	0.957 $\pm$ 0.047	0.1869 $\pm$ 0.0080	0.2516 $\pm$ 0.0117
<i>1 drug seen</i>									
Control	0.439 $\pm$ 0.014	0.042 $\pm$ 0.002	0.659 $\pm$ 0.022	0.0289 $\pm$ 0.0010	0.3192 $\pm$ 0.0115	0.478 $\pm$ 0.016	0.801 $\pm$ 0.029	0.3241 $\pm$ 0.0112	0.8369 $\pm$ 0.0359
PerturbMean	0.664 $\pm$ 0.019	0.062 $\pm$ 0.002	0.728 $\pm$ 0.024	0.0191 $\pm$ 0.0007	-	0.861 $\pm$ 0.031	0.965 $\pm$ 0.034	0.1718 $\pm$ 0.0061	-
MAP	0.692 $\pm$ 0.023	0.137 $\pm$ 0.005	0.741 $\pm$ 0.026	0.0187 $\pm$ 0.0007	0.0950 $\pm$ 0.0034	0.874 $\pm$ 0.031	0.972 $\pm$ 0.035	0.1602 $\pm$ 0.0067	0.2201 $\pm$ 0.0082
<i>0 drug seen</i>									
Control	0.367 $\pm$ 0.014	0.588 $\pm$ 0.024	0.630 $\pm$ 0.025	0.0286 $\pm$ 0.0011	0.4857 $\pm$ 0.0194	0.491 $\pm$ 0.020	0.707 $\pm$ 0.029	0.4495 $\pm$ 0.0179	0.8920 $\pm$ 0.0357
PerturbMean	0.505 $\pm$ 0.018	0.794 $\pm$ 0.029	0.659 $\pm$ 0.023	0.0120 $\pm$ 0.0004	-	0.805 $\pm$ 0.029	0.925 $\pm$ 0.033	0.1646 $\pm$ 0.0059	-
MAP	0.554 $\pm$ 0.019	0.867 $\pm$ 0.030	0.704 $\pm$ 0.025	0.0118 $\pm$ 0.0004	0.0772 $\pm$ 0.0028	0.831 $\pm$ 0.030	0.940 $\pm$ 0.034	0.1490 $\pm$ 0.0054	0.1645 $\pm$ 0.0059

Supplementary Table 14 | Ablation of key hyper-parameters on Tahoe-100M dataset. The ablation factors include the hidden dimension of the Transformer encoder, the weight of expression reconstruction loss term ($\mathcal{L}_{\text{expr}}$) in the loss function, and the number of layers in the Transformer encoder. When ablating each variable, all other factors are kept at their default settings to ensure a controlled comparison. The default values are indicated in the table.

Variant	HVG					Top-50 DEG			
	Pearson Δ $\uparrow$	PDS $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$	Pearson Δ $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$
<i>Hidden layer dimension of Transformer encoder</i>									
512	0.652 $\pm$ 0.012	0.357 $\pm$ 0.007	0.802 $\pm$ 0.018	0.0094 $\pm$ 0.0005	0.1279 $\pm$ 0.0062	0.806 $\pm$ 0.027	0.958 $\pm$ 0.021	0.1205 $\pm$ 0.0039	0.2407 $\pm$ 0.0156
768	0.651 $\pm$ 0.013	0.361 $\pm$ 0.005	0.824 $\pm$ 0.020	0.0092 $\pm$ 0.0004	0.1158 $\pm$ 0.0059	0.793 $\pm$ 0.026	0.946 $\pm$ 0.019	0.1341 $\pm$ 0.0055	0.2358 $\pm$ 0.0147
1024 (default)	0.659 $\pm$ 0.014	0.344 $\pm$ 0.006	0.807 $\pm$ 0.021	0.0097 $\pm$ 0.0004	0.1216 $\pm$ 0.0073	0.822 $\pm$ 0.023	0.934 $\pm$ 0.024	0.1264 $\pm$ 0.0046	0.2299 $\pm$ 0.0138
1280	0.667 $\pm$ 0.011	0.350 $\pm$ 0.006	0.814 $\pm$ 0.025	0.0095 $\pm$ 0.0003	0.1194 $\pm$ 0.0066	0.814 $\pm$ 0.028	0.941 $\pm$ 0.023	0.1306 $\pm$ 0.0049	0.2382 $\pm$ 0.0117
<i>Weight λ of expression reconstruction loss term</i>									
0.05	0.657 $\pm$ 0.015	0.338 $\pm$ 0.006	0.775 $\pm$ 0.023	0.0100 $\pm$ 0.0004	0.1228 $\pm$ 0.0085	0.859 $\pm$ 0.022	0.917 $\pm$ 0.025	0.1297 $\pm$ 0.0044	0.2234 $\pm$ 0.0129
0.1 (default)	0.659 $\pm$ 0.014	0.344 $\pm$ 0.006	0.807 $\pm$ 0.021	0.0097 $\pm$ 0.0004	0.1216 $\pm$ 0.0073	0.822 $\pm$ 0.023	0.934 $\pm$ 0.024	0.1264 $\pm$ 0.0046	0.2299 $\pm$ 0.0138
0.15	0.668 $\pm$ 0.017	0.327 $\pm$ 0.007	0.801 $\pm$ 0.017	0.0102 $\pm$ 0.0005	0.1256 $\pm$ 0.0079	0.834 $\pm$ 0.020	0.919 $\pm$ 0.029	0.1219 $\pm$ 0.0041	0.2193 $\pm$ 0.0161
0.2	0.652 $\pm$ 0.014	0.334 $\pm$ 0.005	0.772 $\pm$ 0.022	0.0099 $\pm$ 0.0005	0.1267 $\pm$ 0.0083	0.856 $\pm$ 0.021	0.904 $\pm$ 0.020	0.1223 $\pm$ 0.0054	0.2215 $\pm$ 0.0132
<i>Number of layers in the Transformer encoder</i>									
2	0.648 $\pm$ 0.016	0.359 $\pm$ 0.007	0.803 $\pm$ 0.019	0.0093 $\pm$ 0.0004	0.1171 $\pm$ 0.0061	0.785 $\pm$ 0.025	0.923 $\pm$ 0.027	0.1354 $\pm$ 0.0038	0.2416 $\pm$ 0.0150
4 (default)	0.659 $\pm$ 0.014	0.344 $\pm$ 0.006	0.807 $\pm$ 0.021	0.0097 $\pm$ 0.0004	0.1216 $\pm$ 0.0073	0.822 $\pm$ 0.023	0.934 $\pm$ 0.024	0.1264 $\pm$ 0.0046	0.2299 $\pm$ 0.0138
6	0.674 $\pm$ 0.016	0.329 $\pm$ 0.005	0.819 $\pm$ 0.024	0.0101 $\pm$ 0.0003	0.1162 $\pm$ 0.0081	0.847 $\pm$ 0.019	0.942 $\pm$ 0.028	0.1328 $\pm$ 0.0052	0.2186 $\pm$ 0.0114

Supplementary Table 15 | Ablation of key hyper-parameters on OP3 dataset. The ablation factors include the hidden dimension of the Transformer encoder, the weight of expression reconstruction loss term ($\mathcal{L}_{\text{expr}}$) in the loss function, and the number of layers in the Transformer encoder. When ablating each variable, all other factors are kept at their default settings to ensure a controlled comparison. The default values are indicated in the table.

Variant	HVG					Top-50 DEG			
	Pearson $\Delta \uparrow$	PDS $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$	Pearson $\Delta \uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$
<i>Hidden layer dimension of Transformer encoder</i>									
512	0.605 $\pm$ 0.019	0.262 $\pm$ 0.010	0.637 $\pm$ 0.017	0.0373 $\pm$ 0.0034	0.2356 $\pm$ 0.0121	0.854 $\pm$ 0.006	0.919 $\pm$ 0.015	0.6475 $\pm$ 0.0618	0.6130 $\pm$ 0.0384
768	0.602 $\pm$ 0.023	0.282 $\pm$ 0.008	0.680 $\pm$ 0.020	0.0207 $\pm$ 0.0028	0.2198 $\pm$ 0.0147	0.849 $\pm$ 0.008	0.932 $\pm$ 0.013	0.5531 $\pm$ 0.0712	0.6116 $\pm$ 0.0329
1024 (default)	0.601 $\pm$ 0.021	0.263 $\pm$ 0.009	0.692 $\pm$ 0.019	0.0288 $\pm$ 0.0031	0.2229 $\pm$ 0.0134	0.861 $\pm$ 0.007	0.921 $\pm$ 0.014	0.6176 $\pm$ 0.0664	0.5986 $\pm$ 0.0359
1280	0.604 $\pm$ 0.020	0.276 $\pm$ 0.009	0.675 $\pm$ 0.018	0.0282 $\pm$ 0.0032	0.2086 $\pm$ 0.0138	0.859 $\pm$ 0.006	0.928 $\pm$ 0.018	0.5802 $\pm$ 0.0574	0.5871 $\pm$ 0.0401
<i>Weight λ of expression reconstruction loss term</i>									
0.05	0.596 $\pm$ 0.025	0.253 $\pm$ 0.008	0.665 $\pm$ 0.016	0.0282 $\pm$ 0.0034	0.2327 $\pm$ 0.0120	0.869 $\pm$ 0.007	0.898 $\pm$ 0.013	0.6414 $\pm$ 0.0567	0.5776 $\pm$ 0.0388
0.1 (default)	0.601 $\pm$ 0.021	0.263 $\pm$ 0.009	0.692 $\pm$ 0.019	0.0288 $\pm$ 0.0031	0.2229 $\pm$ 0.0134	0.861 $\pm$ 0.007	0.921 $\pm$ 0.014	0.6176 $\pm$ 0.0664	0.5986 $\pm$ 0.0359
0.15	0.608 $\pm$ 0.018	0.251 $\pm$ 0.010	0.694 $\pm$ 0.022	0.0301 $\pm$ 0.0027	0.2124 $\pm$ 0.0156	0.852 $\pm$ 0.008	0.914 $\pm$ 0.010	0.5893 $\pm$ 0.0771	0.6217 $\pm$ 0.0314
0.2	0.605 $\pm$ 0.020	0.262 $\pm$ 0.009	0.671 $\pm$ 0.018	0.0276 $\pm$ 0.0032	0.2361 $\pm$ 0.0155	0.868 $\pm$ 0.006	0.923 $\pm$ 0.016	0.6616 $\pm$ 0.0574	0.5837 $\pm$ 0.0401
<i>Number of layers in the Transformer encoder</i>									
2	0.586 $\pm$ 0.025	0.259 $\pm$ 0.008	0.659 $\pm$ 0.016	0.0279 $\pm$ 0.0035	0.2335 $\pm$ 0.0127	0.853 $\pm$ 0.006	0.884 $\pm$ 0.017	0.6357 $\pm$ 0.0599	0.5819 $\pm$ 0.0326
4 (default)	0.601 $\pm$ 0.021	0.263 $\pm$ 0.009	0.692 $\pm$ 0.019	0.0288 $\pm$ 0.0031	0.2229 $\pm$ 0.0134	0.861 $\pm$ 0.007	0.921 $\pm$ 0.014	0.6176 $\pm$ 0.0664	0.5986 $\pm$ 0.0359
6	0.602 $\pm$ 0.017	0.264 $\pm$ 0.011	0.708 $\pm$ 0.023	0.0297 $\pm$ 0.0029	0.2193 $\pm$ 0.0136	0.871 $\pm$ 0.008	0.939 $\pm$ 0.016	0.5996 $\pm$ 0.0742	0.6153 $\pm$ 0.0305

Supplementary Table 16 | Ablation of key hyper-parameters on SciPlex3 dataset. The ablation factors include the hidden dimension of the Transformer encoder, the weight of expression reconstruction loss term ($\mathcal{L}_{\text{expr}}$) in the loss function, and the number of layers in the Transformer encoder. When ablating each variable, all other factors are kept at their default settings to ensure a controlled comparison. The default values are indicated in the table.

Variant	HVG					Top-50 DEG			
	Pearson $\Delta \uparrow$	PDS $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$	Pearson $\Delta \uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$
<i>Hidden layer dimension of Transformer encoder</i>									
512	0.684 $\pm$ 0.009	0.266 $\pm$ 0.008	0.828 $\pm$ 0.004	0.0286 $\pm$ 0.0020	0.0142 $\pm$ 0.0010	0.842 $\pm$ 0.009	0.898 $\pm$ 0.011	0.1912 $\pm$ 0.0101	0.0610 $\pm$ 0.0042
768	0.679 $\pm$ 0.011	0.286 $\pm$ 0.006	0.865 $\pm$ 0.005	0.0307 $\pm$ 0.0017	0.0127 $\pm$ 0.0008	0.812 $\pm$ 0.011	0.927 $\pm$ 0.009	0.1765 $\pm$ 0.0083	0.0665 $\pm$ 0.0035
1024 (default)	0.698 $\pm$ 0.010	0.275 $\pm$ 0.007	0.848 $\pm$ 0.004	0.0294 $\pm$ 0.0018	0.0135 $\pm$ 0.0009	0.827 $\pm$ 0.010	0.913 $\pm$ 0.010	0.1844 $\pm$ 0.0092	0.0632 $\pm$ 0.0038
1280	0.702 $\pm$ 0.010	0.263 $\pm$ 0.007	0.834 $\pm$ 0.004	0.0279 $\pm$ 0.0019	0.0146 $\pm$ 0.0010	0.839 $\pm$ 0.009	0.905 $\pm$ 0.011	0.1938 $\pm$ 0.0099	0.0649 $\pm$ 0.0041
<i>Weight λ of expression reconstruction loss term</i>									
0.05	0.676 $\pm$ 0.009	0.284 $\pm$ 0.006	0.862 $\pm$ 0.004	0.0309 $\pm$ 0.0016	0.0124 $\pm$ 0.0008	0.808 $\pm$ 0.009	0.930 $\pm$ 0.008	0.1785 $\pm$ 0.0080	0.0615 $\pm$ 0.0033
0.1 (default)	0.698 $\pm$ 0.010	0.275 $\pm$ 0.007	0.848 $\pm$ 0.004	0.0294 $\pm$ 0.0018	0.0135 $\pm$ 0.0009	0.827 $\pm$ 0.010	0.913 $\pm$ 0.010	0.1844 $\pm$ 0.0092	0.0632 $\pm$ 0.0038
0.15	0.705 $\pm$ 0.012	0.268 $\pm$ 0.008	0.834 $\pm$ 0.005	0.0287 $\pm$ 0.0020	0.0143 $\pm$ 0.0011	0.841 $\pm$ 0.011	0.906 $\pm$ 0.012	0.1920 $\pm$ 0.0105	0.0648 $\pm$ 0.0044
0.2	0.699 $\pm$ 0.011	0.259 $\pm$ 0.007	0.842 $\pm$ 0.006	0.0278 $\pm$ 0.0021	0.0139 $\pm$ 0.0010	0.846 $\pm$ 0.012	0.899 $\pm$ 0.011	0.1967 $\pm$ 0.0109	0.0669 $\pm$ 0.0042
<i>Number of layers in the Transformer encoder</i>									
2	0.682 $\pm$ 0.011	0.287 $\pm$ 0.006	0.829 $\pm$ 0.004	0.0306 $\pm$ 0.0020	0.0141 $\pm$ 0.0010	0.809 $\pm$ 0.009	0.918 $\pm$ 0.011	0.1768 $\pm$ 0.0085	0.0661 $\pm$ 0.0041
4 (default)	0.698 $\pm$ 0.010	0.275 $\pm$ 0.007	0.848 $\pm$ 0.004	0.0294 $\pm$ 0.0018	0.0135 $\pm$ 0.0009	0.827 $\pm$ 0.010	0.913 $\pm$ 0.010	0.1844 $\pm$ 0.0092	0.0632 $\pm$ 0.0038
6	0.703 $\pm$ 0.009	0.262 $\pm$ 0.008	0.866 $\pm$ 0.005	0.0281 $\pm$ 0.0017	0.0128 $\pm$ 0.0008	0.842 $\pm$ 0.011	0.921 $\pm$ 0.009	0.1932 $\pm$ 0.0103	0.0607 $\pm$ 0.0035

Supplementary Table 17 | Ablation study on representation integration. The ablation variants include: (1) **w/o cell emb**, where the cell embedding (CLS token) derived from STATE SE-600M is replaced with a learnable parameter; (2) **w/o SE & KG gene emb**, where the fused gene tokens are excluded from the input sequence of the perturbation model; (3) **w/o SE gene emb**, where the gene tokens produced by STATE SE-600M are removed; and (4) **w/o KG gene emb**, where the pre-trained gene embeddings are removed.

Variant	HVG					Top-50 DEG			
	Pearson $\Delta \uparrow$	PDS $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$	Pearson $\Delta \uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$
<i>OP3 (unseen drug)</i>									
w/o cell emb	0.267 $\pm$ 0.023	0.232 $\pm$ 0.008	0.578 $\pm$ 0.018	0.0661 $\pm$ 0.0034	1.2754 $\pm$ 0.0148	0.583 $\pm$ 0.007	0.788 $\pm$ 0.015	1.0857 $\pm$ 0.0599	1.9375 $\pm$ 0.0328
w/o SE & KG gene emb	0.514 $\pm$ 0.019	0.248 $\pm$ 0.010	0.616 $\pm$ 0.021	0.0431 $\pm$ 0.0029	0.4379 $\pm$ 0.0152	0.698 $\pm$ 0.008	0.866 $\pm$ 0.013	0.9412 $\pm$ 0.0712	1.1293 $\pm$ 0.0395
w/o SE gene emb	0.567 $\pm$ 0.022	0.241 $\pm$ 0.009	0.620 $\pm$ 0.017	0.0297 $\pm$ 0.0032	0.2730 $\pm$ 0.0126	0.824 $\pm$ 0.006	0.900 $\pm$ 0.016	0.7641 $\pm$ 0.0628	0.7138 $\pm$ 0.0371
w/o KG gene emb	0.584 $\pm$ 0.020	0.245 $\pm$ 0.010	0.612 $\pm$ 0.018	0.0278 $\pm$ 0.0030	0.2853 $\pm$ 0.0149	0.848 $\pm$ 0.007	0.905 $\pm$ 0.014	0.7519 $\pm$ 0.0685	0.7027 $\pm$ 0.0349
MAP	0.601 $\pm$ 0.021	0.263 $\pm$ 0.009	0.692 $\pm$ 0.019	0.0288 $\pm$ 0.0031	0.2229 $\pm$ 0.0134	0.861 $\pm$ 0.007	0.921 $\pm$ 0.014	0.6176 $\pm$ 0.0664	0.5986 $\pm$ 0.0359
<i>SciPlex3 (unseen drug)</i>									
w/o cell emb	0.342 $\pm$ 0.012	0.141 $\pm$ 0.008	0.787 $\pm$ 0.003	0.0423 $\pm$ 0.0018	0.0264 $\pm$ 0.0009	0.638 $\pm$ 0.012	0.730 $\pm$ 0.008	0.3359 $\pm$ 0.0116	0.1329 $\pm$ 0.0040
w/o SE & KG gene emb	0.559 $\pm$ 0.008	0.144 $\pm$ 0.005	0.811 $\pm$ 0.003	0.0357 $\pm$ 0.0021	0.0192 $\pm$ 0.0012	0.693 $\pm$ 0.011	0.822 $\pm$ 0.010	0.2415 $\pm$ 0.0108	0.1137 $\pm$ 0.0047
w/o SE gene emb	0.612 $\pm$ 0.009	0.178 $\pm$ 0.006	0.813 $\pm$ 0.004	0.0324 $\pm$ 0.0016	0.0173 $\pm$ 0.0007	0.778 $\pm$ 0.010	0.893 $\pm$ 0.010	0.2067 $\pm$ 0.0082	0.0748 $\pm$ 0.0036
w/o KG gene emb	0.607 $\pm$ 0.007	0.136 $\pm$ 0.006	0.829 $\pm$ 0.005	0.0308 $\pm$ 0.0013	0.0155 $\pm$ 0.0007	0.735 $\pm$ 0.010	0.887 $\pm$ 0.011	0.2174 $\pm$ 0.0076	0.0814 $\pm$ 0.0033
MAP	0.698 $\pm$ 0.010	0.275 $\pm$ 0.007	0.848 $\pm$ 0.004	0.0294 $\pm$ 0.0018	0.0135 $\pm$ 0.0009	0.827 $\pm$ 0.010	0.913 $\pm$ 0.010	0.1844 $\pm$ 0.0092	0.0632 $\pm$ 0.0038
<i>OP3 (unseen cell type-drug combination)</i>									
w/o cell emb	0.237 $\pm$ 0.046	0.114 $\pm$ 0.003	0.571 $\pm$ 0.021	0.0502 $\pm$ 0.0706	0.3199 $\pm$ 0.0068	0.524 $\pm$ 0.013	0.753 $\pm$ 0.012	0.6415 $\pm$ 0.0593	0.4499 $\pm$ 0.0387
w/o SE & KG gene emb	0.522 $\pm$ 0.019	0.142 $\pm$ 0.007	0.627 $\pm$ 0.011	0.0430 $\pm$ 0.0334	0.2178 $\pm$ 0.0175	0.660 $\pm$ 0.010	0.868 $\pm$ 0.029	0.4197 $\pm$ 0.0658	0.3543 $\pm$ 0.0192
w/o SE gene emb	0.568 $\pm$ 0.053	0.177 $\pm$ 0.005	0.638 $\pm$ 0.030	0.0134 $\pm$ 0.0811	0.2164 $\pm$ 0.0129	0.785 $\pm$ 0.027	0.934 $\pm$ 0.019	0.2939 $\pm$ 0.1076	0.2972 $\pm$ 0.0451
w/o KG gene emb	0.569 $\pm$ 0.024	0.167 $\pm$ 0.008	0.626 $\pm$ 0.018	0.0122 $\pm$ 0.0527	0.1930 $\pm$ 0.0154	0.774 $\pm$ 0.016	0.938 $\pm$ 0.033	0.2932 $\pm$ 0.0984	0.2778 $\pm$ 0.0265
MAP	0.595 $\pm$ 0.028	0.223 $\pm$ 0.004	0.663 $\pm$ 0.016	0.0107 $\pm$ 0.0419	0.1638 $\pm$ 0.0093	0.815 $\pm$ 0.014	0.957 $\pm$ 0.017	0.2763 $\pm$ 0.0857	0.2476 $\pm$ 0.0234

Supplementary Table 18 | Cell-line-specific ablation study on the knowledge-pretrained gene embedding. The best-performing and worst-performing cell lines of OP3 (T cells CD4+, Myeloid cells) and SciPlex3 (K562, A549) datasets are selected as representative examples. For the **w/o KG gene emb**, the pre-trained gene embeddings are removed.

Variant	HVG					Top-50 DEG			
	Pearson $\Delta \uparrow$	PDS $\uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$	Pearson $\Delta \uparrow$	Direction Acc $\uparrow$	MSE $\downarrow$	Wasserstein $\downarrow$
<i>OP3 - T cells CD4+</i>									
w/o KG gene emb	0.670 $\pm$ 0.019	0.224 $\pm$ 0.010	0.714 $\pm$ 0.015	0.0663 $\pm$ 0.0020	0.2240 $\pm$ 0.0191	0.904 $\pm$ 0.019	0.955 $\pm$ 0.014	0.3857 $\pm$ 0.0253	0.6284 $\pm$ 0.0299
MAP	0.678 $\pm$ 0.021	0.240 $\pm$ 0.005	0.732 $\pm$ 0.021	0.0647 $\pm$ 0.0024	0.1925 $\pm$ 0.0173	0.912 $\pm$ 0.031	0.991 $\pm$ 0.015	0.3724 $\pm$ 0.0296	0.4738 $\pm$ 0.0356
<i>OP3 - Myeloid cells</i>									
w/o KG gene emb	0.527 $\pm$ 0.023	0.128 $\pm$ 0.012	0.701 $\pm$ 0.011	0.0632 $\pm$ 0.0026	0.2855 $\pm$ 0.0138	0.833 $\pm$ 0.021	0.948 $\pm$ 0.017	0.3656 $\pm$ 0.0304	0.7426 $\pm$ 0.0247
MAP	0.588 $\pm$ 0.028	0.194 $\pm$ 0.006	0.718 $\pm$ 0.014	0.0538 $\pm$ 0.0017	0.2673 $\pm$ 0.0216	0.864 $\pm$ 0.035	0.970 $\pm$ 0.012	0.2843 $\pm$ 0.0392	0.6372 $\pm$ 0.0471
<i>SciPlex3 - K562</i>									
w/o KG gene emb	0.632 $\pm$ 0.011	0.146 $\pm$ 0.005	0.833 $\pm$ 0.018	0.0124 $\pm$ 0.0005	0.0124 $\pm$ 0.0014	0.823 $\pm$ 0.012	0.932 $\pm$ 0.025	0.1312 $\pm$ 0.0098	0.0589 $\pm$ 0.0031
MAP	0.682 $\pm$ 0.013	0.180 $\pm$ 0.007	0.842 $\pm$ 0.009	0.0103 $\pm$ 0.0004	0.0110 $\pm$ 0.0023	0.864 $\pm$ 0.015	0.960 $\pm$ 0.016	0.1114 $\pm$ 0.0082	0.0397 $\pm$ 0.0033
<i>SciPlex3 - A549</i>									
w/o KG gene emb	0.601 $\pm$ 0.014	0.121 $\pm$ 0.006	0.867 $\pm$ 0.017	0.0120 $\pm$ 0.0004	0.0183 $\pm$ 0.0010	0.760 $\pm$ 0.015	0.905 $\pm$ 0.018	0.1972 $\pm$ 0.0125	0.0844 $\pm$ 0.0023
MAP	0.645 $\pm$ 0.010	0.146 $\pm$ 0.008	0.880 $\pm$ 0.007	0.0113 $\pm$ 0.0005	0.0142 $\pm$ 0.0027	0.795 $\pm$ 0.011	0.941 $\pm$ 0.020	0.1634 $\pm$ 0.0106	0.0651 $\pm$ 0.0041

Supplementary Table 19 | Runing time scalability analysis of the knowledge-driven multi-modal pre-training. Convergence time (hours) and normalized training time (hours per %) across different training data proportions are shown.

Training data proportion	25%	50%	75%	100%
Convergence time (h)	6.72	13.19	19.92	25.44
Normalized time (h/%)	0.269	0.264	0.266	0.254

Supplementary Table 20 | Comparison of running time and memory usage across methods. The comparison was conducted on the Tahoe-100M, under the zero-shot generalization setting for unprofiled drugs. All methods were trained with the same pseudo-bulk size of 24 using four A100 GPUs. We report the convergence time, defined as the training time required to trigger the early-stopping criterion after 1,000 steps without a new minimum loss, and the peak memory usage per GPU.

Methods	MAP	XPert	CRISP	chemCPA	PRnet
Converge time (h)	11.91	12.25	8.63	19.90	6.77
Peak Memory Usage (GB)	42.48	28.54	20.15	9.86	16.71

A.6 GSEA Analysis and Interpretation Criteria

In Section 2.4, we performed pathway-level functional evaluation based on GSEA to assess both *in silico* drug screening performance and gene set-level mechanism consistency. Below, we further clarify the GSEA analysis pipeline and the criteria used to interpret enrichment results and derive biological conclusions.

GSEA methodology. We performed GSEA using `gseapy.gsea` [49], a standard GSEA tool, on perturbed expression profiles against matched control profiles. For each drug, the predicted and control expression matrices over the 2,000 HVGs were formatted as genes-by-samples matrices and compared using phenotype permutation with 1,000 permutations. Gene sets were provided in GMT format from MSigDB curated pathway collections. We used the signal-to-noise statistic for gene ranking, with a minimum gene-set size of 10. Pathways were considered significant if they satisfied $FDR < 0.05$ and $|NES| > 1.0$. This pipeline was consistently applied to both predicted and true expression profiles. GSEA was performed within the dataset-specific top-2,000 HVG universe, matching the modeled output space of all perturbation predictors. This avoids introducing genes not directly predicted by the models and provides a more stable readout for sparse single-cell perturbation data. The reported NES and FDR values should therefore be interpreted as HVG-restricted enrichment statistics.

GSEA result interpretation criteria. For simulated *in silico* drug screening, we focus on a curated set of disease-relevant pathways for A-549, defined based on established cancer-associated signaling programs. Following [19], for each drug, we compute an aggregate pathway score by summing the absolute normalized enrichment scores ($|NES|$) across these pathways. Only significant pathways ($FDR < 0.05$) are retained. As these pathways correspond to disease-driving programs expected to be suppressed, larger absolute enrichment reflects stronger predicted therapeutic potential. Drugs are ranked accordingly, and performance is evaluated using top- k recall against approved anti-cancer drugs.

For gene set-level mechanism agreement, conclusions are drawn by comparing predicted and ground-truth enrichment patterns using consistent criteria. Agreement is assessed in terms of both enrichment direction (sign of NES) and relative magnitude across pathways. For each drug, we consider pathways that are biologically relevant to its mechanism of action and expected effects in NSCLC, based on established drug-target annotations (mainly DrugBank [50]) and disease-associated pathways. Supplementary Table S10 reports these pathway-level comparisons, supporting the consistency between predicted and known mechanisms.

A.7 Literature support for homoharringtonine in A-549 screening

We performed literature mining for the top-ranked non-standard candidates from the A549 simulated screening. Among these compounds, homoharringtonine (HHT; omacetaxine mepesuccinate) emerged as one of the most strongly supported candidates with direct A549 evidence. In a study of NSCLC cell lines, HHT showed cytotoxic activity against A549 cells, with a reported IC_{50} of $3.7 \mu M$, and inhibited A549 cell proliferation, reduced viable cell numbers, and suppressed clonogenic growth in a time- and dose-dependent manner. Mechanistically, the same study showed that HHT suppressed STAT3 signaling in A549 cells, including inhibition of STAT3 protein phosphorylation and blockade of IL-6-induced STAT3 activation and nuclear translocation, suggesting that its predicted activity is consistent with perturbation of a cancer-relevant survival pathway [34]. In addition, an independent study in *KRAS*-mutant lung adenocarcinoma models reported that HHT inhibited A549 proliferation in vitro and further supported its anti-tumor relevance in *KRAS*-driven lung adenocarcinoma contexts [35]. A further mechanistic study proposed the TMEM16A channel as a lung-cancer-relevant target of HHT, showing TMEM16A protein expression in A549 cells and HHT-mediated inhibition of TMEM16A channel currents, apoptosis induction, and tumor suppression in an LA795 xenograft model [36]. Together, these A549-focused findings support HHT as a promising literature-backed candidate among MAP’s top-ranked non-standard compounds. This analysis strengthens the evidence that MAP can nominate potentially meaningful drug leads beyond simply retrieving already approved therapies.

A.8 Detailed functional analysis on A-549

In Section 2.4, we performed pathway-level evaluation by applying GSEA to MAP’s zero-shot predictions on held-out drugs in A549, and showed that the resulting ranking prioritizes four clinically established NSCLC

therapies. Here we complement this ranking with drug-by-drug functional interpretations by inspecting representative gene sets (pathways) that are linked to each compound’s known mechanism of action and to disease-relevant programs in A549.

Afatinib. Afatinib is an irreversible pan-*ERBB* inhibitor [51]. In Figure 4c, MAP closely matches the experimental GSEA on inflammatory and stress-associated programs, including NF- κ B signaling-related pathways, with highly concordant NES values. This agreement is biologically plausible given the prominent role of NF- κ B-mediated survival and inflammatory transcription in lung adenocarcinoma contexts [52, 53], and indicates that the model captures treatment-associated remodeling of these core programs. In addition, MAP reproduces a modest activation of apoptosis-related pathways, consistent with drug-induced cellular stress responses observed in the measured perturbation profiles.

Crizotinib. Crizotinib is a clinically used kinase inhibitor in NSCLC [54]. Figure 4c shows that MAP accurately recovers the enrichment of NRF2 pathway-associated oxidative stress response pathways, consistent with the well-characterized KEAP1-NRF2 antioxidant-defense axis in A549 [55]. The model also matches the enrichment of cell-cycle-related programs, indicating that it preserves proliferative regulatory states reflected in the transcriptomic data. Together, the concurrent agreement on NRF2 pathway and cell-cycle gene sets supports that MAP yields context-consistent and mechanistically interpretable pathway readouts for this compound.

Adagrasib. Adagrasib is a covalent inhibitor targeting the mutant KRAS^{G12C} protein and is designed to attenuate downstream RAS-MAPK signaling [56, 57]. In the A549 perturbation data, key KRAS pathway-associated oncogenic programs exhibit clear negative enrichment, and MAP closely reproduces these suppressive patterns. Notably, the model correctly predicts downregulation of KRAS-driven lung cancer signatures, accompanied by reduced activity of downstream proliferative programs such as MYC-associated gene sets. This coordinated agreement forms a coherent pathway-level response that is consistent with attenuation of KRAS-linked transcriptional output observed in the experimental GSEA.

Osimertinib. Osimertinib is a third-generation EGFR tyrosine kinase inhibitor [58]. In Figure 4c, MAP reproduces the experimentally observed activation of an EGFR signaling response signature, consistent with acute pathway perturbation and compensatory/feedback regulation captured in the transcriptomic profiles. At the same time, the model matches persistent enrichment of KRAS-related oncogenic programs and NF- κ B-associated inflammatory pathways, indicating that it correctly preserves dominant disease-relevant transcriptional programs in this cellular context [59, 60]. Overall, these results demonstrate that MAP recovers drug-proximal responses while maintaining biologically coherent background regulation at the gene-set level.

A.9 Evaluation Metrics

Pearson delta correlation calculates Pearson correlation coefficient (PCC) between predicted and observed log fold changes (logFC) in gene expression, and quantifies how well the model recovers the magnitude of perturbation-induced expression changes. Formally, given a perturbed pseudo-bulk $x^{(p)}$, the model prediction $\hat{x}^{(p)}$ and a paired control pseudo-bulk x , the Pearson Delta Correlation is calculated as

$$Pr\Delta = PCC(\log FC(\bar{x}^{(p)}, \bar{x}), \log FC(\bar{\hat{x}}^{(p)}, \bar{x})) \quad (13)$$

where $\bar{\hat{x}}$, $\bar{p}$ and $\bar{c}$ denote the cell-wise mean expression of corresponding pseudo-bulks, and $PCC(\cdot, \cdot)$ is calculated as:

$$PCC(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (14)$$

The coefficient ranges from -1 to 1, where 1 indicates perfect positive correlation, -1 indicates perfect negative correlation, and 0 indicates no linear relationship. We calculate $Pr\Delta$ for both highly variable genes (HVG) and top-50 differentially expressed (DE) genes.

Direction accuracy evaluates whether the model predicts the correct direction of regulation relative to the

control state, and is formulated as:

$$\text{Acc}(\bar{x}^{(p)}, \hat{\bar{x}}^{(p)}, \bar{x}) = \frac{1}{N} \sum_{i=1}^N \mathbb{1}\left(\text{sign}(\bar{x}_i^{(p)} - \bar{x}_i) = \text{sign}(\hat{\bar{x}}_i^{(p)} - \bar{x}_i)\right), \quad (15)$$

where N is the number of evaluated genes (HVGs or top-50 DE genes), $x_i \in \mathbb{R}$ denotes the expression value of i -th gene, $\text{sign}(\cdot)$ denotes the sign of the input value and $\mathbb{1}(\cdot)$ denotes the indicator function

Perturbation discrimination score assesses whether the predicted perturbation effects are distinguishable across different perturbations. Given T perturbations, we define for each perturbation p :

$$r_{\text{norm}}^p = \frac{1}{T} \sum_{k \neq p} \mathbb{1}\left(d(\hat{\bar{x}}^{(p)}, \bar{x}^{(k)}) < d(\hat{\bar{x}}^{(p)}, \bar{x}^{(p)})\right), \quad (16)$$

where $d(\cdot, \cdot)$ is the Manhattan distance. The perturbation discrimination score is defined as:

$$\text{PDS} = 1 - \frac{2}{T} \sum_{t=1}^T r_{\text{norm}}^p. \quad (17)$$

We calculate perturbation discrimination score only on HVGs to ensure consistency across perturbations.

Mean squared error (MSE) quantifies the absolute deviation between predicted and observed expression values at the gene level. Unlike Pearson delta correlation, which evaluates relative agreement in variation patterns, MSE directly measures the magnitude of prediction error in the original value space. It is defined as:

$$\text{MSE}(\bar{x}^{(p)}, \hat{\bar{x}}^{(p)}) = \frac{1}{N} \sum_{i=1}^N \left(\hat{\bar{x}}_i^{(p)} - \bar{x}_i^{(p)}\right)^2, \quad (18)$$

where N is the number of evaluated genes (HVGs or top-50 DE genes), and $\bar{x}_i^{(p)}$ and $\hat{\bar{x}}_i^{(p)}$ denote the observed and predicted expression values of the i -th gene under perturbation p , respectively. Lower MSE indicates more accurate prediction of perturbation-induced expression magnitudes.

Wasserstein distance evaluates the discrepancy between predicted and observed perturbation responses at the distribution level, capturing differences beyond pointwise errors. The metric measures the minimal cost of transforming the predicted expression distribution into the observed one, thereby reflecting global structural differences in expression profiles. Formally, let P and Q denote the empirical distributions of predicted and observed gene expression values under perturbation p , respectively. The Wasserstein distance is defined as:

$$W(P, Q) = \inf_{\gamma \in \Pi(P, Q)} \int_{\mathcal{X} \times \mathcal{Y}} |x - y|, d\gamma(x, y), \quad (19)$$

where $\Pi(P, Q)$ denotes the set of all joint distributions with marginals P and Q . In practice, we compute the Wasserstein distance between $\hat{\bar{x}}^{(p)}$ and $\bar{x}^{(p)}$ over the selected gene set (HVGs or top-50 DE genes). Lower values indicate better alignment between predicted and observed expression distributions.

A.10 Details of Knowledge Sources

To ensure the diversity, high quality and broad coverage of MAP-KG, we curate and integrate fragmented knowledge from 14 public sources. Here, we briefly introduce each source below, and summarize their contributions of knowledge data in Table 21.

BindingDB [61] is a public database specializing in experimentally measured binding affinities of small molecules to their protein targets.

BioGRID [62] (Biological General Repository for Interaction Datasets) is a curated biological database of protein, chemical, and genetic interactions. It archives interaction data from multiple organisms, supporting network-based analysis of cellular pathways and gene function.

ChEMBL [63] is a manually curated, open-source database of bioactive molecules with drug-like properties, maintained by the European Bioinformatics Institute (EMBL-EBI). It provides a vast repository of high-quality bioactivity data extracted from scientific literature and deposited datasets.

DrugBank [50] is a bioinformatics and cheminformatics resource that combines detailed drug data with comprehensive drug target information. It integrates chemical, pharmacological, and pharmaceutical data with detailed target sequence and pathway information, making it a gold-standard database for pharmacologists and pharmaceutical researchers.

DrugCentral [64] is a comprehensive online drug database that provides information about active pharmaceutical ingredients. It serves as a key resource for drug discovery and development by integrating drug data from various authoritative sources, including information on indications and pharmacological actions.

Ensembl [40] is a comprehensive genome database that provides integrated access to high-quality annotated genome sequences for vertebrates and other model organisms. It aggregates genomic and gene annotation data, including gene models, transcript and protein sequences, variation and comparative genomics information, and provides tools for data retrieval and analysis to support a range of genomic research tasks.

HGNC [41] (HUGO Gene Nomenclature Committee) database is the authoritative resource for standardized human gene nomenclature. It assigns unique symbols and names to each human gene to ensure clear scientific communication and facilitate unambiguous data retrieval from publications and other databases.

Human PPI [65] is a computational resource that provides a predicted human protein-protein interactome. It allows users to search for interactions via gene names or UniProt identifiers and to check for interactions between specific protein pairs.

IntAct [66] is an open-source database system that provides a freely available resource for molecular interaction data. It offers a rich platform for studying protein functions in cellular processes through experimentally determined interactions.

PrimeKG [38] is a biomedical knowledge graph that integrates curated information from public resources into a unified graph. It links entities such as diseases, drugs, genes/proteins, pathways/biological processes, phenotypes, and anatomical concepts. We incorporate the drug-gene relationships from this resource into MAP-KG.

PubChem [39] is a public repository of chemical molecules and their biological activities, maintained by the National Center for Biotechnology Information (NCBI). It provides extensive data on chemical structures, identifiers, physicochemical properties, and bioassay results, serving as a foundational resource for chemical biology and drug discovery research.

STITCH [67] (Search Tool for Interacting Chemicals) is a database of known and predicted interactions between chemicals and proteins. It integrates interaction data from experiments, databases, and literature to build comprehensive networks.

STRING [68] is a biological database and web resource for known and predicted protein-protein interactions. These interactions include both direct (physical) and indirect (functional) associations.

TTD [69] (Therapeutic Target Database) is a specialized resource that provides information about known therapeutic protein and nucleic acid targets, along with their targeted diseases, pathway information and the

corresponding drugs directed at each of these targets.

Supplementary Table 21 | Links and statistics of the 14 knowledge sources in MAP-KG .

Source	Link	Type of Feature	Count
ChEMBL [63]	https://www.ebi.ac.uk/chembl	Drug	16,002
DrugBank [50]	https://www.drugbank.com	Drug	17,459
PubChem [39]	https://pubchem.ncbi.nlm.nih.gov	Drug	168,209
Ensembl [40]	https://www.ensembl.org	Gene	86,599
HGNC [41]	https://www.genenames.org/	Gene	44,433
BindingDB [61]	https://www.bindingdb.org	Drug-Gene	279,857
DrugCentral [64]	https://drugcentral.org	Drug-Gene	28,665
STITCH [67]	http://stitch.embl.de	Drug-Gene	81,346
TTD [69]	https://db.idrblab.net/ttd	Drug-Gene	32,920
PrimeKG [38]	https://zitniklab.hms.harvard.edu/projects/PrimeKG/	Drug-Gene	23904
BioGRID [62]	https://thebiogrid.org	Drug-Gene	5,085
		Gene-Gene	219,214
HumanPPI [65]	http://prodata.swmed.edu/humanPPI	Gene-Gene	24,178
IntAct [66]	https://www.ebi.ac.uk/intact	Gene-Gene	10,623
STRING [68]	https://string-db.org	Gene-Gene	20,881

Supplementary Table 22 | Representative examples of nodes and edges in MAP-KG. We list example drug nodes, gene nodes, and relation edges with the attributes used in our graph construction. Long text fields are truncated to the first 25–40 words for readability. Canonical gene sequences are included as attributes in MAP-KG but omitted here due to length.

Drug nodes.			
PubChem ID	Drug name	SMILES	MOA (truncated)
10013505	NGX-267	<chem>CC[C@@H]1SC2(CCN(C)CC2)NC1=O</chem>	NGX267 has been shown to stimulate M1 receptors in a fashion analogous to acetylcholine, a neuro...
100154	CA4P	<chem>COC1=C(C=C(C=C1)CC(C2=CC(=C(C(=C2)OC)OC)OC)OC)O)O</chem>	CA4P binds tubulin with a higher efficacy than colchicines, and was therefore initially investigated as...
10026	Amyl Nitrite	<chem>CCCCCON=O</chem>	Amyl nitrite’s antianginal action is thought to be the result of a reduction in systemic and pulmonary...

Gene nodes.			
Gene stable ID	Gene name	Gene summary (truncated)	Function annotation (truncated)
ENSG00000198888	<i>MT-ND1</i>	mitochondrially encoded NADH: ubiquinone oxidoreductase core subunit 1; Enables NADH dehydrogenase (ubiquinone)...	FUNCTION: Core subunit of the mitochondrial membrane respiratory chain NADH dehydrogenase (Complex I) which catalyzes...
ENSG00000291382	<i>CPLANE2</i>	ciliogenesis and planar polarity effector complex subunit 2	FUNCTION: Required for efficient primary cilia initiation, regulating a late step in cilia initiation. Plays a role in the final...
ENSG00000165246	<i>NLGN4Y</i>	neuroligin 4 Y-linked; This gene encodes a type I membrane protein that belongs to the family of neuroligins...	FUNCTION: Cell surface protein involved in cell-cell-interactions via its interactions with neuroligin family members...

Drug–gene relations.		
Drug PubChem ID	Gene stable ID	Relation text
44230999	ENSG00000125384	functions as an agonist on the protein target
65243	ENSG00000145888	functions as an inhibitor on the protein target
4956	ENSG00000127955	functions as a reactant involved in biochemical processes of the protein encoded by gene

Gene–gene relations.		
Gene 1 stable ID	Gene 2 stable ID	Relation text
ENSG00000123159	ENSG00000196586	is in the same protein complex or physically associated with
ENSG00000166033	ENSG00000174951	is in the same protein complex or physically associated with
ENSG00000130713	ENSG00000075914	encodes a protein that engages with high confidence (1.0) in a protein–protein interaction with the protein encoded by

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